NATIONAL INSTITUTE OF TECHNOLGY PATNA
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
MID-SEMESTER EXAMINATION – OCTOBER, 2024
B. Tech CSE/ B.Tech-M.Tech-DD-CSE (CS/DS)/ Minor in AI
CS54118/CS5439– Machine Learning
V[th] Semester Time:2 hours Max.Marks:30
Answer All questions

Q. No	Question	Marks	CO	BL
1	a. Identify the kind of machine learning task/ type of machine learning (in some cases, two or more tasks can be combined to address the problem) that can be applied to the following applications and justify your answer: i) An online banking system manages transactions, storing the user's past transaction history, including whether the transaction is fraudulent or not. Verify whether the given new transaction is fraudulent or not. ii) A biologist collects DNA sequence data; in this data, some patients' disease names have been given, and the rest of the patients' disease names have not been given. Which DNA mutations are disease-causing and which are not? iii) There is a cart to which a pole is attached, and it is unstable as the centre mass of the pole is above the pivot point. The player should move the cart left or right to balance it. If the pole deviates more than 15 degrees, the game ends. Otherwise, for every timestamp (move) the pole deviates no more than 15 degrees, one point is rewarded to the agent. iv) A broker stores the details of a house and the price of the house. Predict the price of a house given the details of a house. v) Actuaries (persons whose job involves calculating insurance risks and payments for insurance companies) at health insurance companies are often used to identify groups of consumers that use their health insurance in specific ways. vi) Scientists might use different amounts of fertilizer and water in different fields to see how they affect crop yield.	9M	CO1, CO2, CO5	Remember, Understand, Analyze
	b. Can we use a linear regression algorithm to solve the Binary classification problem? Justify your answer.	1M	CO1	Remember, Understand

2	a. Which of the following characteristics best suits a dataset that is used in the Naïve Bayes Classifier and why? i) Fewer number of observations ii) Huge number of observations iii) Zero frequency of an attribute value iv) Feature Independence	2M	CO1, CO5	Understar Evaluate
	b. Consider the following example with two independent attributes, brightness and quantity and a dependent variable, price.	8M	CO5	Understand Apply

Brightness	Quantity	Price
3	5	38
5	8	51
8	1	38
9	3	55
4	0	25
2	6	33
8	7	71
6	4	51

Apply the multiple linear regression model and compute the updated parameters of the model up to *two* iterations (epochs) by using the learning rate = 0.001 and with the following initial random values (in your computation, roundup each value up to two digits): *Intercept* (β_0) = *1.50*, *Slope of Brightness* (β_1) = *0.5*, *Slope of Quantity* (β_2) = *0.1*

3	a. Identify the limitation of ***Information Gain*** to use as a splitting criterion in the decision tree algorithm. Suggest a solution to overcome this limitation.	2M	CO1, CO5	Evaluate, Create
	b. Consider the following training examples: (Profit is the target variable)	8M	CO1, CO5	Understand, Apply

Age	Category	Type	Profit
Old	Employee	Grade-A	No
Old	Business	Grade-A	Yes
Old	Business	Grade-B	No
Mid	Employee	Grade-A	No
Mid	Employee	Grade-B	No
Mid	Business	Grade-B	Yes
Mid	Business	Grade-A	Yes

Construct the Decision Tree using an ID3-based Decision Tree Algorithm, and draw the final tree by showing the splitting attribute.

********** ALL THE BEST **********

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END-SEMESTER EXAMINATION – DECEMBER, 2024

B. Tech CSE/ B.Tech-M.Tech-DD-CSE (CS/DS)/ Minor in AI

CS54118/CS5439– Machine Learning

Vth Semester Time:3 hours Max.Marks:60

Answer All questions

Q. No	Question	Marks CO, BL
1	a. An ANN is proposed to predict the results of soccer matches. The designer has information about the two competing teams and the conditions of the match and is supposed to predict how many goals each team will score. Let us say that every team consists of 20 players and the following information are available. • The skill level of every player on each of the two teams. Skill is rated by a group of soccer reporters on a scale from 0 ("is unable to kick the ball") to 10 ("world class player"). • The number of matches that each team has played during the last two weeks. There are never more than seven matches in that period of time. • The statistics of former matches between the same two teams within the past 10 years (e.g., Team A won 30% of the matches, Team B 45%, and 25% of the matches were tied). • The continent that each team comes from (North America, South America, Europe, Africa, Asia, or Australia). • Where the match takes place (Team A's stadium, Team B's stadium, or neutral place). • The phase of the soccer season (early season vs. late season). Find the number of **input neuron, number of hidden layers, number of output neurons.** Draw the structure of the network. What learning algorithm can be used? Properly justify your answer.	07M CO5 CO10, CO11 Understand Apply Create
	b. When do we prefer logistic regression over other machine learning algorithms? Give an idea on how we can use Logistic Regression for multi-class classification. Explain overfitting. Explain the purpose of validation dataset in preventing overfitting.	08M CO5 Understand, Evaluate
2	a. Consider the architecture of the CNN as follows: ***Input size:*** $32 \times 32 \times 3$; **Conv layer**: 16 filters and each filter size 3×3, stride is one, padding is one; **Pooling Layer**: Max-pooling 2×2, and stride is two. **Fully connected layer**: 256 neurons. **Output Layer**: 8 neurons. Compute the output size and number of parameters after each layer and the total parameters in the network. Assume that biases are also included in the network.	07M CO5 Remember, Evaluate
	b. Why does SVM focus on maximizing the margin between classes instead of minimizing classification errors directly? How does this help in generalization? What is kernel trick and how it is useful in SVM?	03M CO5, CO6 Understand
	c. Define reinforcement learning. How does it differ from supervised and unsupervised learning? Provide real-world examples to illustrate the differences. Also, discuss about the key components of reinforcement learning system: ***agent, environment, state, action, reward, and policy***.	05M CO9 Remember Understand

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3	a. The data tuples are sorted by decreasing probability values which are returned by a logistic regression (LR) for a binary classification task, and is shown in the following table. Compute the Confusion matrix, Precession, Recall, Accuracy and F1-score. (Mention any assumptions made by you). **Tuple Number / True Class / Probability (Given LR)** 1 / P / 0.95 2 / N / 0.85 3 / P / 0.78 4 / P / 0.66 5 / N / 0.60 6 / P / 0.55 7 / N / 0.53 8 / N / 0.52 9 / N / 0.45 10 / P / 0.40	7M CO1 CO5 Remember Understand Apply
	b. Define and differentiate bagging and boosting in ensemble learning, also discuss in detail about both of their working procedure. Describe how **Random Forest** build its trees differently from traditional decision trees.	8M CO5 Understand Create
4	a. Compute the agglomerative hierarchical clustering using single-linkage approach on the following six data points in 2-D space: **Data Point / Data Point in 2-D space** A / (1, 1) B / (1.5, 1.5) C / (5, 5) D / (3, 4) E / (4, 4) F / (3, 3.5) Use *Manhattan distance measure* to find the distance between two data points, and also show the clustering process using ***dendrogram***.	07M CO3 Apply
	b. Mention the need of dimensionality reduction methods. Assume that the we are given the dataset with four data points in 2-D space: {(2, 4), (3, 5), (5, 7), (7, 9)}. Perform the Principal Component Analysis to reduce the dataset from 2-D to 1-D (what is the 1-D representation of the data after projection onto the principal component), and how much variance retained in 1-D representation?	8M CO4 CO3 Understand Apply Analyze

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